Solving Systems by Elimination with Scaling
Answer Key

Algebra 1   Linear Systems and Modeling

Quick Answers

1. $x = 3$, $y = 5$
2. $x = 2$, $y = 1$
3. $a = 7$, $c = 5$
4. $x = 2$, $y = 1$
5. $x = 2$, $y = $ -3
6. $m = 28$, $s = 22$
7. $x = 3$, $y = $ -1
8. $b = 10$, $c = 2$
9. $x = 6$, $y = 2$
10. $x = 5$, $y = 15$

Curriculum

Level: Algebra 1   Linear Systems and Modeling
HSA-REI.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 2–5 of 5 across 11 tagged checks.

1. School store: $2x + y = 11$, $x + 3y = 18$.
Target. The y coefficients are 1 and 3, so scaling the first equation by 3 matches them — cheaper than matching x.

$$\begin{array}{rcl} 3(2x + y) = 3(11) & \implies & 6x + 3y = 33 \\ \text{subtract } x + 3y = 18 & \implies & 5x = 15, \quad x = 3 \end{array}$$

Back-substitute into $2x + y = 11$: $6 + y = 11$, so $y = 5$. Check in the second equation: $3 + 3(5) = 18$ ✓
A notebook costs \$3 and a binder costs \$5. $\boxed{(x, y) = (3, 5)}$

2. $4x - 3y = 5$, $x + y = 3$.
Target. Scale the second equation by 3 so the y terms are $-3y$ and $+3y$.

$$\begin{array}{rcl} 3(x + y) = 3(3) & \implies & 3x + 3y = 9 \\ \text{add to } 4x - 3y = 5 & \implies & 7x = 14, \quad x = 2 \end{array}$$

Back-substitute into $x + y = 3$: $y = 1$. Check: $4(2) - 3(1) = 5$ ✓ $\boxed{(x, y) = (2, 1)}$

3. Theater tickets: $a + c = 12$, $9a + 5c = 88$.
Target. Scale the count equation by 5 to match the $5c$ term.

$$\begin{array}{rcl} 5(a + c) = 5(12) & \implies & 5a + 5c = 60 \\ \text{subtract from } 9a + 5c = 88 & \implies & 4a = 28, \quad a = 7 \end{array}$$

Then $c = 12 - 7 = 5$. Check the money equation: $9(7) + 5(5) = 63 + 25 = 88$ ✓
The theater sold 7 adult tickets and 5 child tickets. $\boxed{(a, c) = (7, 5)}$

4. $3x + 4y = 10$, $5x + 6y = 16$.
Target. No coefficient divides another, so scale *both*. For x, the least common multiple of 3 and 5 is 15.

$$\begin{aligned} 5(3x+4y) &= 5(10) &\Longrightarrow&& 15x+20y &= 50 \\ 3(5x+6y) &= 3(16) &\Longrightarrow&& 15x+18y &= 48 \end{aligned}$$

Subtract: $2y = 2$, so $y = 1$. Back-substitute into $3x + 4y = 10$: $3x = 6$, $x = 2$. Check: $5(2) + 6(1) = 16$ ✓ $\boxed{(x, y) = (2, 1)}$

5. $5x + 2y = 4$, $3x + y = 3$.
Target. The y coefficients are 2 and 1, so scale the second equation by 2.

$$\begin{aligned} 2(3x+y) &= 2(3) &\Longrightarrow&& 6x+2y &= 6 \\ \text{subtract } 5x+2y &= 4 &\Longrightarrow&& x &= 2 \end{aligned}$$

Back-substitute into $3x + y = 3$: $6 + y = 3$, so $y = -3$. Check the first equation: $5(2) + 2(-3) = 10 - 6 = 4$ ✓ $\boxed{(x, y) = (2, -3)}$

6. Bakery: $m + s = 50$, $3m + 4s = 172$.
Target. Scale the count equation by 4 to match the $4s$ term.

$$\begin{aligned} 4(m+s) &= 4(50) &\Longrightarrow&& 4m+4s &= 200 \\ \text{subtract from } 3m+4s &= 172 &\Longrightarrow&& -m &= -28, \quad m = 28 \end{aligned}$$

Then $s = 50 - 28 = 22$. Check the money equation: $3(28) + 4(22) = 84 + 88 = 172$ ✓
The bakery sold 28 muffins and 22 scones. $\boxed{(m, s) = (28, 22)}$

7. $4x + 5y = 7$, $6x + 7y = 11$.
Target. Both need scaling. Eliminating x is cheaper: the least common multiple of 4 and 6 is 12, while y would need 35.

$$\begin{aligned} 3(4x+5y) &= 3(7) &\Longrightarrow&& 12x+15y &= 21 \\ 2(6x+7y) &= 2(11) &\Longrightarrow&& 12x+14y &= 22 \end{aligned}$$

Subtract: $y = -1$. Back-substitute into $4x + 5y = 7$: $4x - 5 = 7$, so $x = 3$. Check: $6(3) + 7(-1) = 18 - 7 = 11$ ✓ $\boxed{(x, y) = (3, -1)}$

8. Canoe: $2b + 2c = 24$, $3b - 3c = 24$.
Target. The c coefficients are 2 and -3; their least common multiple is 6, so scale both.

$$\begin{aligned} 3(2b+2c) &= 3(24) &\Longrightarrow&& 6b+6c &= 72 \\ 2(3b-3c) &= 2(24) &\Longrightarrow&& 6b-6c &= 48 \end{aligned}$$

Add: $12b = 120$, so $b = 10$. Back-substitute into $2b+2c = 24$: $20+2c = 24$, so $c = 2$. Check the upstream equation: $3(10) - 3(2) = 24$ ✓
The canoe moves at 10 kilometres per hour in still water and the current runs at 2 kilometres per hour. $\boxed{(b, c) = (10, 2)}$

9. Maya's mistake on $x + 4y = 14$, $3x - 2y = 14$.

(a) What is wrong. Maya multiplied only the left side by 3 and left the right side at 14. That is a different equation, not the same one rewritten: $3x + 12y = 14$ has different solutions from $x + 4y = 14$. Multiplying *every* term — both sides — by 3 is allowed because it scales both sides of a true statement equally, so exactly the same pairs (x, y) still satisfy it. *This part is flagged for manual review; a student answer that says "she has to multiply the* 14 *too, or the equation is no longer equivalent" earns full credit.*

(b) The correct work.

$$\begin{aligned} 3(x + 4y) = 3(14) \quad &\Longrightarrow \quad 3x + 12y = 42 \\ \text{subtract } 3x - 2y = 14 \quad &\Longrightarrow \quad 14y = 28, \quad y = 2 \end{aligned}$$

Back-substitute into $x + 4y = 14$: $x + 8 = 14$, so $x = 6$. Check: $3(6) - 2(2) = 18 - 4 = 14$ ✓ $\boxed{(x, y) = (6, 2)}$

10. Salt mixture: $x + y = 20$, $30x + 50y = 45 \cdot 20$.

Setting up the second equation. Percent times litres is the salt in each part, and the salt in the two parts must equal the salt in the mixture: $45 \cdot 20 = 900$.

Target. Scale the volume equation by 30 to match the $30x$ term.

$$\begin{aligned} 30(x + y) = 30(20) \quad &\Longrightarrow \quad 30x + 30y = 600 \\ \text{subtract from } 30x + 50y = 900 \quad &\Longrightarrow \quad 20y = 300, \quad y = 15 \end{aligned}$$

Then $x = 20 - 15 = 5$. Check the salt equation: $30(5) + 50(15) = 150 + 750 = 900$ ✓

The technician uses 5 litres of the 30% solution and 15 litres of the 50% solution.

$\boxed{(x, y) = (5, 15)}$